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EDITED AND REVIEWED BY

Jingzheng Ren,
Hong Kong Polytechnic University, Hong
Kong SAR, China

*CORRESPONDENCE

Adam C. G. Cooper
✉ adam.cooper@ucl.ac.uk

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Editorial: Transdisciplinary engineering for sustainability decisions

Adam C. G. Cooper^{1*}, Bryan Moser² and Susan Krumdieck³

¹Department of Science Technology Engineering and Public Policy, University College London, London, United Kingdom, ²Systems Design and Management, Massachusetts Institute of Technology, Cambridge, MA, United States, ³The Island Centre for Net Zero, School of Energy Geoscience Infrastructure and Society, Heriot-Watt University, Edinburgh, United Kingdom

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Editorial on the Research Topic

Transdisciplinary engineering for sustainability decisions

The sustainability challenges countries around the world are facing demand thoughtful action—insight that draws across a range of expertise, and integrates it for maximum impact. In addition to the notion of integrating expertise, is the often overlooked role of engineering—the skill of doing and making. Sustainability challenges are as much a challenge of doing and making—what we do and what we make—as it is a challenge of policy and regulation, that is, decision making. That is why, with this Research Topic we wanted to draw attention to the special role of engineering in sustainability, recognizing it plays a role both at the level of developing technologies that might mitigate or avoid environmental harm, but also contribute to the very process of government policy making that supports that engineering activity. This Research Topic then looks at policy and practice decision-making, in government, communities and industry for sustainability. It looks at the special role of engineering within that context. And being in that context, it means that transdisciplinary engineering practice needs to be foregrounded. Transdisciplinarity typically captures the idea of bringing together different disciplinary knowledge experts together with non-disciplinary knowledge experts to make better outcomes. Transdisciplinary engineering centers engineering in this process—recognizing the special role engineering has in shaping societal outcomes. In so doing, It forces reflection on what kind of transdisciplinarity is needed in engineering to get the sustainability outcomes we need in society? Here 6 papers explore different ways that transdisciplinarity with engineering can manifest for sustainability policy and decision-making processes.

Each contributor addresses the topic from different perspectives and in different contexts but all with common underlying ideas. They cover energy (in both domestic use and industrial production), transport (from both a business and aging perspectives), electrical waste and agricultural processes.

[Boulton and Krumdieck](#) explore the potential for the Systems Transitions Engineering approach to leverage change in international oil companies in their paper “Transition Engineering co-design sprint: oil company business model.” This leverages core engineering practices and concepts (“sprints” and “systems”) generate attention beyond simply the technical sphere and into the decision-making spaces

of oil companies for more sustainable future business models. This shows how engineering, when brought into new transdisciplinary practices can enable changes even in the places where change seems least likely to happen.

Hu et al. look at the improved management of methane emissions from farm ruminants using a systems model and decision-support software in their paper “Transdisciplinary model-based systems engineering in the development of the Ruminant Farm Systems model.” Like Boulton and Krumdieck, they address a sector responsible for significant climate emissions, and do so with a systems-based approach as well. What differs here of course is the focus less on reshaping an entire business model and more to refining a business model so that the fundamental practices continue but in a more sustainable way.

Ismail Mohamed et al. investigate the management of a different kind of waste—electronic. Their paper “A sustainable approach tackling WEEE management using ontology-based DSS” moves us away from the climate crisis in sustainability terms toward wider environmental pollutants and possibilities of material recovery. They develop the transdisciplinary engineering field of computational sustainability—leveraging the concept of ontologies to support integration across the diverse industries connect to electronic waste. This again illustrates the potential for software systems to enable better sustainability but, to gain those benefits, significant integration of perspectives and concepts (i.e., transdisciplinarity) is often needed.

Ojima et al. research the potential for making transport options better both for sustainability and for older people, recognizing that any improved system that works only classic sustainability metrics (e.g., reduced emissions) won’t be sustainable if older people, in an aging population, can’t or won’t use it. Their paper “A cost-effective strategy for enhancing mobility in aging communities: the case of Narita City” reflects a core transdisciplinary approach—ensuring that we do not solve problems only on one dimension but on multiple. They also rely on software modeling but move away from systems-based approaches into a clear model that captures multiple dimensions of Quality of Mobility.

Trigos and Osorio present an analysis similar to Ojima et al., aiming to develop a process that enables firms to better route their vehicles for customer deliveries. This is an equally important issue as mobility for older people as logistics becomes ever more complex and threatening not just environmental sustainability but social sustainability and worker welfare. In their paper “Integrating the Triple Bottom Line into the Vehicle Routing Problem: a transdisciplinary approach to customer prioritization” they develop an approach that parallels the work of Ismail Mohamed et al. insofar as it seeks to connect concepts of what good (or useful) is to business decisions more directly.

Wise et al. look at how systems methods need to be combined in a government setting with transdisciplinary practice to ensure the full potential of transdisciplinary engineering is released. In their paper “The potential for a transdisciplinary systems approach to improve national policy analysis: learning from UK cases of home energy transitions” they use historic and contemporary case studies to illustrate why this approach is both different from current

(failing) practice and reflects implicit effects of effective systems approaches from (successful) historic case. As with as with Boulton and Krumdieck and Hu et al., systems concepts are central.

Each of these papers take the concept and practice of transdisciplinary engineering—understood in different ways and deploys it into business or government contexts to address real world problems like climate and energy, mobility and transport, agricultural and electronic waste management. For some, systems approaches are key, others digital models or platforms serve as sites of either knowledge values integration in order to ensure decisions about resources are effective, sustainable and positive for society. We hope this Research Topic provides a platform for future transdisciplinary engineering development and brings more scholars and practitioners to the International Society of Transdisciplinary Engineering, who sponsored and led this Research Topic.

Author contributions

AC: Conceptualization, Project administration, Writing – original draft, Writing – review & editing. BM: Conceptualization, Data curation, Project administration, Validation, Writing – review & editing. SK: Conceptualization, Data curation, Project administration, Validation, Writing – review & editing.

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